

# William G. Underwood

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## Employment

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- Assistant Professor, Department of Statistics** 2026–  
*University of Warwick*
- Postdoctoral Research Associate in Statistics** 2024–2026  
*University of Cambridge*
- Statistical Laboratory, DPMMS
  - Advisor: Richard Samworth
- Assistant in Instruction** 2020–2024  
*Princeton University*

## Education

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- PhD in Operations Research & Financial Engineering** 2019–2024  
*Princeton University*
- Department of Operations Research & Financial Engineering
  - Dissertation: Estimation and Inference in Modern Nonparametric Statistics
  - Advisor: Matias Cattaneo
- MA in Operations Research & Financial Engineering** 2019–2021  
*Princeton University*
- MMath in Mathematics & Statistics** 2015–2019  
*University of Oxford, St John's College*

## Research & publications

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### Articles

- Efficient and minimax optimal in-context nonparametric regression with transformers, with M. Ching, I. Popescu, N. Smith, T. Ma and R. J. Samworth. *International Conference on Machine Learning*, forthcoming, 2026. [arXiv:2601.15014](#).
- Inference with Mondrian random forests, with M. D. Cattaneo and J. M. Klusowski. *Journal of the Royal Statistical Society, Series B* 88(3), 1060–1085, 2026. [arXiv:2310.09702](#).
- Sharp anti-concentration inequalities for extremum statistics via copulas, with M. D. Cattaneo and R. P. Masini. *Bernoulli* 32(3), 2022–2045, 2026. [arXiv:2502.07699](#).
- Yurinskii's coupling for martingales, with M. D. Cattaneo and R. P. Masini. *Annals of Statistics* 53(5), 2179–2203, 2025. [arXiv:2210.00362](#).
- Uniform inference for kernel density estimators with dyadic data, with M. D. Cattaneo and Y. Feng. *Journal of the American Statistical Association* 119(548), 2695–2708, 2024. [arXiv:2201.05967](#).
- Motif-based spectral clustering of weighted directed networks, with A. Elliott and M. Cucuringu. *Applied Network Science* 5(62), 2020. [arXiv:2004.01293](#).
- Simple Poisson PCA: an algorithm for (sparse) feature extraction with simultaneous dimension determination, with L. Smallman and A. Artemiou. *Computational Statistics* 35, 559–577, 2019.

### Preprints

- Upgrading survival models with CARE, with H. W. J. Reeve, O. Y. Feng, S. A. Lambert, B. Mukherjee and R. J. Samworth. *Preprint*, 2025. [arXiv:2506.23870](#).

## Working papers

- Higher-order Yurinskii coupling, with M. D. Cattaneo and R. P. Masini.
- Optimal polyadic estimation.

## Presentations & conferences

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|------------------------------------------------------------------------------|----------|
| • Statistical Sciences Research Institute Seminar, University of Southampton | Oct 2026 |
| • Royal Statistical Society International Conference, Bournemouth            | Sep 2026 |
| • International Conference on Statistics and Data Science, Seville           | Dec 2025 |
| • GSEM Statistics Seminar, Université de Genève                              | Nov 2025 |
| • MRC Biostatistics Unit Armitage Workshop, University of Cambridge          | Oct 2025 |
| • Economics Seminar, University of York                                      | Oct 2025 |
| • MRC Biostatistics Unit Seminar, University of Cambridge                    | Sep 2025 |
| • StatMathAppli, Fréjus                                                      | Sep 2025 |
| • London Symposium on Information Theory, Cambridge                          | May 2025 |
| • International Conference on Statistics and Data Science, Nice              | Dec 2024 |
| • Statistics Seminar, University of Pittsburgh                               | Feb 2024 |
| • Statistics Seminar, University of Illinois Urbana-Champaign                | Jan 2024 |
| • Statistics Seminar, University of Michigan                                 | Jan 2024 |
| • Eighth Princeton Day of Statistics, Princeton University                   | Nov 2023 |
| • PhD Poster Session, Two Sigma Investments, New York                        | Jul 2023 |
| • Statistical Foundations of Data Science, Princeton University              | May 2023 |
| • Research Symposium, Two Sigma Investments, New York                        | Jun 2022 |
| • Seventh Princeton Day of Statistics, Princeton University                  | Oct 2021 |
| • Statistics Laboratory, Princeton University                                | Sep 2021 |

## Software

- care-survival: upgrading survival models with CARE in Python, 2025.  
GitHub: [wgunderwood/care-survival](https://github.com/wgunderwood/care-survival)
- tex-fmt: LaTeX formatter written in Rust, 2024. GitHub: [wgunderwood/tex-fmt](https://github.com/wgunderwood/tex-fmt)
- MondrianForests: Mondrian random forests in Julia, 2023.  
GitHub: [wgunderwood/MondrianForests.jl](https://github.com/wgunderwood/MondrianForests.jl)
- DyadicKDE: dyadic kernel density estimation in Julia, 2022.  
GitHub: [wgunderwood/DyadicKDE.jl](https://github.com/wgunderwood/DyadicKDE.jl)
- motifcluster: motif-based spectral clustering in R, Python and Julia, 2020.  
GitHub: [wgunderwood/motifcluster](https://github.com/wgunderwood/motifcluster)

## Awards & funding

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|----------------------------------------------------------------------------------------|------|
| • Early Career Researcher Grant, G-Research                                            | 2025 |
| • School of Engineering and Applied Science Award for Excellence, Princeton University | 2022 |
| • Francis Robbins Upton Fellowship in Engineering, Princeton University                | 2019 |
| • Royal Statistical Society Prize, Royal Statistical Society & University of Oxford    | 2019 |
| • Gibbs Statistics Prize, University of Oxford                                         | 2019 |
| • James Fund for Mathematics Research Grant, St John's College, University of Oxford   | 2017 |
| • Casberd Scholarship, St John's College, University of Oxford                         | 2016 |

## Teaching

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### Lecturer

*University of Warwick*

- ST343: Topics in Data Science: Causal Inference Spring 2027
- ST925: Graduate Topics in Statistics: Statistical Theory of Neural Networks Spring 2027
- ST921: Statistical Frontiers: Nonparametric Inference via Resampling Autumn 2026

### Lecturer

*University of Cambridge*

- Part III Statistical Learning in Practice Lent 2026

### Supervisor

*University of Cambridge*

- Part III Essay on Score Matching and Diffusion Models Lent 2026
- Part II Probability & Measure Michaelmas 2025
- Part IB Statistics Lent 2025
- Part III Essay on Inference with Random Forests Lent 2025

### Drop-in Session Leader

*University of Cambridge*

- Part III Modern Statistical Methods Lent 2025
- Part III Topics in Statistical Theory Lent 2025
- Part III Concentration Inequalities Lent 2025

### Assistant in Instruction

*Princeton University*

- ORF 499: Senior Thesis Spring 2024
- ORF 498: Senior Independent Research Foundations Fall 2023
- SML 201: Introduction to Data Science Fall 2023
- ORF 363: Computing and Optimization Spring 2023
- ORF 524: Statistical Theory and Methods Fall 2022
- ORF 526: Probability Theory Fall 2022
- ORF 524: Statistical Theory and Methods Fall 2021
- ORF 245: Fundamentals of Statistics Spring 2021
- ORF 363: Computing and Optimization Fall 2020

## Professional experience

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**Quantitative Research Intern** 2023

*Two Sigma Investments*

**Machine Learning Consultant** 2018

*Mercury Digital Assets*

**Educational Consultant** 2018

*Polaris & Dawn*

**Premium Tutor** 2016–2018

*MyTutor*

**Statistics and Machine Learning Researcher** 2017–2017

*Cardiff University*

**Data Science Intern** 2017

*Rolls-Royce*

## Service

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### Peer review

*Annals of the Institute of Statistical Mathematics; Annals of Statistics; Bernoulli; Biometrika; Econometric Theory; Information Theory, Probability and Statistical Learning; International Conference on Machine Learning; Journal of the American Statistical Association; Journal of Business & Economic Statistics; Journal of Causal Inference; Journal of Econometrics; Journal of Machine Learning Research; Journal of Nonparametric Statistics; Journal of the Royal Statistical Society, Series B; Journal of Statistical Computation and Simulation; Neural Information Processing Systems; Operations Research; SIAM Journal on Mathematics of Data Science; Statistical Science; Statistics and Computing; Technometrics.*

### Other service

- Organiser, Statistical Learning & Inference Seminar, University of Warwick 2026–2027
- Organiser, Statistics Department Conference, University of Warwick 2026–2027
- Consultant, Cambridge Statistics Clinic, University of Cambridge 2024–2026
- Local organiser, Statistical Foundations of Data Science, Princeton University 2023

## References

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- Richard Samworth, Professor, Statistical Laboratory, DPMMS, University of Cambridge
- Matias Cattaneo, Professor, Department of Economics, Princeton University
- Jianqing Fan, Professor, ORFE, Princeton University
- Jason Klusowski, Associate Professor, ORFE, Princeton University